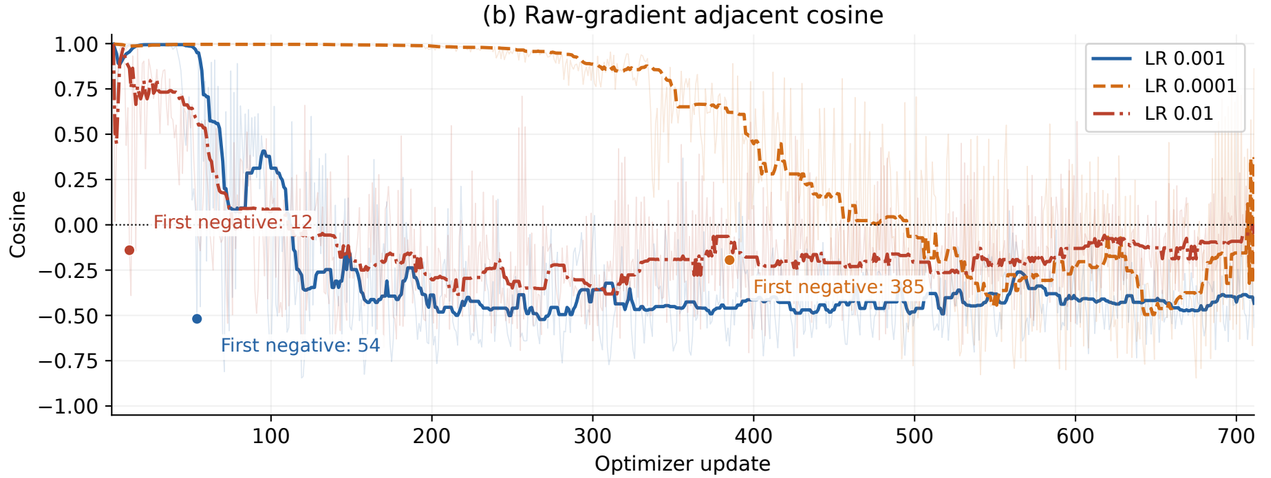
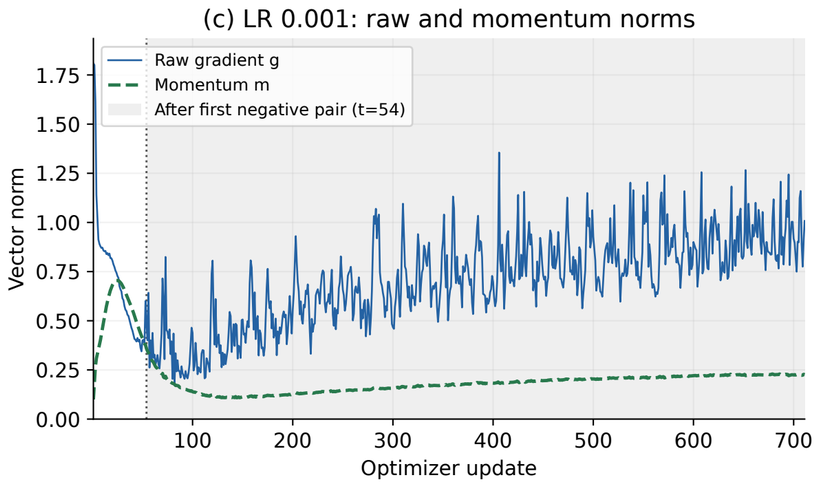
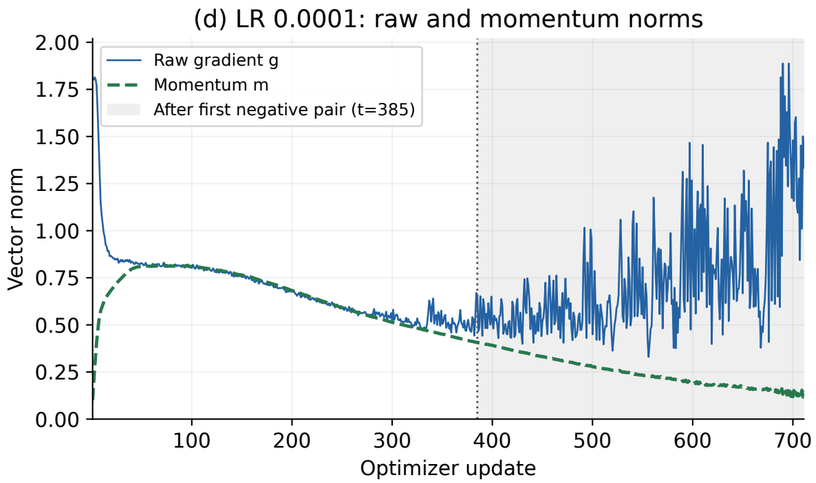
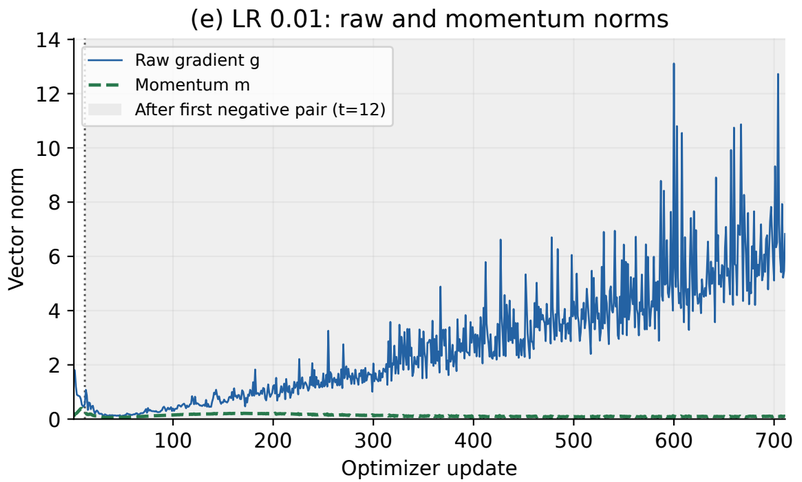
Pythia-14M: loss, raw-gradient alignment and vector norms
Three completed warmup-only LR runs; matched initial weights and data order; seed 1234











Light traces: individual updates. Bold loss: trailing 25-update token-weighted mean; bold cosine: trailing 25-update median.
Dots/vertical markers: first negative unsmoothed raw pair. Shading marks subsequent time, including later positive pairs.
Norm curves are unsmoothed; each lower panel has its own linear y-scale. g is pre-clipping; m is uncorrected exp_avg from clipped gradients.
Display: updates 1-711; partial update 712 retained in results. The sign crossing does not define the onset of norm separation.